

# Agentic AI for Information Retrieval

A Multi-Agent System for Multi-Hop Retrieval with  
Self-Reflective Validation

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## **Abstract**

# Chapter 1

## Introduction to Agentic AI in Information Retrieval

1.1 From Reactive Generation to Agency

1.2 The Limits of Static Ranking

1.3 Problem Statement

1.4 Objectives and Contributions

1.5 Structure of This Document

## Chapter 2

# Data and Preprocessing

### 2.1 Requirements for Evaluating an Agentic IR System

### 2.2 On the Datasets Listed in the Assignment Brief

### 2.3 HotpotQA

### 2.4 2WikiMultihopQA

### 2.5 Preprocessing Pipeline

### 2.6 Evaluation Sampling

Table 2.1: Corpus statistics after preprocessing, read from `{dataset}_corpus_stats.json`. Passages are de-duplicated by content under a canonical title-derived id; sentence splits are the datasets’ own and are persisted verbatim, because gold supporting facts index into them by position and re-splitting anywhere downstream would renumber the ground truth while every metric still looked plausible. - marks a statistic the build did not record.

| Statistic                     | hotpotqa | twowiki |
|-------------------------------|----------|---------|
| Questions in split            | 7 405    | 12 576  |
| Paragraphs read               | 73 700   | 125 760 |
| Passages after de-duplication | 66 581   | 54 957  |
| Identical paragraphs dropped  | 7 119    | 68 186  |
| Title collisions resolved     | 0        | 2 617   |
| Sentences                     | 269 602  | 202 759 |
| Mean sentences per passage    | 4.05     | 3.69    |
| Mean passage length (chars)   | 531.9    | 427.2   |
| Mean sentence length (chars)  | 130.6    | 115.1   |
| Questions with gold triples   | 0        | 12 576  |
| Gold supporting facts         | 18 005   | 30 687  |
| Supporting facts unresolvable | 1        | 0       |
| Evaluation sample             | 250      | 250     |

Table 2.2: Composition of the frozen evaluation slice against the full split. The slice is stratified on question type and sampled once, with seed 42, by `scripts/sample_eval_set.py`; its SHA-256 is recorded in every run’s `meta.json` and it is never re-sampled at run time. HotpotQA’s validation split is entirely `level=hard`, so difficulty gives one stratum and a silently unstratified sample; question type is the informative split there.

| Question type     | Split $n$ | Split share | Sample $n$ | Sample share |
|-------------------|-----------|-------------|------------|--------------|
| <i>hotpotqa</i>   |           |             |            |              |
| bridge            | 5 918     | 0.799       | 200        | 0.800        |
| comparison        | 1 487     | 0.201       | 50         | 0.200        |
| <i>total</i>      | 7 405     | 1.000       | 250        | 1.000        |
| <i>twowiki</i>    |           |             |            |              |
| bridge_comparison | 2 751     | 0.219       | 55         | 0.220        |
| comparison        | 3 040     | 0.242       | 60         | 0.240        |
| compositional     | 5 236     | 0.416       | 104        | 0.416        |
| inference         | 1 549     | 0.123       | 31         | 0.124        |
| <i>total</i>      | 12 576    | 1.000       | 250        | 1.000        |

## Chapter 3

# Retrieval Methodologies

### 3.1 Classical Foundations

#### 3.1.1 Sparse Retrieval: BM25

#### 3.1.2 Dense Retrieval

#### 3.1.3 Hybrid Fusion and Reranking

### 3.2 Why Single-Shot Retrieval Fails on Multi-Hop Needs

### 3.3 Agentic Retrieval

#### 3.3.1 Query Decomposition and Refinement

#### 3.3.2 Tool-Augmented Retrieval

#### 3.3.3 Knowledge Graph Traversal

#### 3.3.4 Self-Reflective Validation

### 3.4 The Role of the Language Model

### 3.5 Architectural Summary



## Chapter 4

# System Implementation and Evaluation

### 4.1 System Architecture

### 4.2 Implementation Details

#### 4.2.1 Indexing

#### 4.2.2 Local Inference

#### 4.2.3 Budget Enforcement

### 4.3 Experimental Setup

### 4.4 Baselines

### 4.5 Evaluation Metrics

#### 4.5.1 Retrieval Quality

#### 4.5.2 Answer Quality

#### 4.5.3 Agent-Specific Measures

### 4.6 Results

### 4.7 Ablation Study

### 4.8 Error Analysis

### 4.9 Discussion

Table 4.1: Answer quality on `hotpotqa`. Exact match and token-level F1 follow the official HotpotQA evaluation script, including its yes/no short circuit; SP-EM and SP-F1 are set metrics over  $(title, sent\_id)$  supporting-fact pairs.  $n$  is the number of questions scored. No `hybrid_rerank` run is available, so no significance marks are shown. - marks a configuration that has not been run.

| System                              | $n$ | EM    | F1    | F1 95% CI      | SP-EM | SP-F1 | $\Delta$ F1 vs. ref. [95% CI] |
|-------------------------------------|-----|-------|-------|----------------|-------|-------|-------------------------------|
| <i>Baselines (weakest first)</i>    |     |       |       |                |       |       |                               |
| <code>bm25_only</code>              | -   | -     | -     | -              | -     | -     | -                             |
| <code>dense_only</code>             | -   | -     | -     | -              | -     | -     | -                             |
| <code>hybrid_rerank</code>          | -   | -     | -     | -              | -     | -     | -                             |
| <code>naive_rag</code>              | -   | -     | -     | -              | -     | -     | -                             |
| <code>self_ask</code>               | -   | -     | -     | -              | -     | -     | -                             |
| <i>Agentic system and ablations</i> |     |       |       |                |       |       |                               |
| <code>agentic_full</code>           | 5   | 0.400 | 0.400 | [0.000, 0.800] | 0.200 | 0.433 | -                             |
| <code>agentic_no_planner</code>     | -   | -     | -     | -              | -     | -     | -                             |
| <code>agentic_no_kg</code>          | -   | -     | -     | -              | -     | -     | -                             |
| <code>agentic_no_verifier</code>    | -   | -     | -     | -              | -     | -     | -                             |

Table 4.2: Retrieval quality on `hotpotqa`, computed with `pytrec_eval` over the reciprocal-rank fusion of every sub-query’s ranking and judged against the gold supporting-fact documents. A multi-hop system issues several queries, so “the ranking” is not directly observed; fusing them is how the orchestrator itself pools evidence, and it needs no score calibration between BM25 and cosine. No `hybrid_rerank` run is available, so no significance marks are shown. - marks a configuration that has not been run.

| System                              | R@2   | R@5   | R@10  | nDCG@10 | nDCG@10 95% CI | MRR   |
|-------------------------------------|-------|-------|-------|---------|----------------|-------|
| <i>Baselines (weakest first)</i>    |       |       |       |         |                |       |
| <code>bm25_only</code>              | -     | -     | -     | -       | -              | -     |
| <code>dense_only</code>             | -     | -     | -     | -       | -              | -     |
| <code>hybrid_rerank</code>          | -     | -     | -     | -       | -              | -     |
| <code>naive_rag</code>              | -     | -     | -     | -       | -              | -     |
| <code>self_ask</code>               | -     | -     | -     | -       | -              | -     |
| <i>Agentic system and ablations</i> |       |       |       |         |                |       |
| <code>agentic_full</code>           | 0.800 | 0.800 | 0.800 | 0.845   | [0.691, 1.000] | 1.000 |
| <code>agentic_no_planner</code>     | -     | -     | -     | -       | -              | -     |
| <code>agentic_no_kg</code>          | -     | -     | -     | -       | -              | -     |
| <code>agentic_no_verifier</code>    | -     | -     | -     | -       | -              | -     |

Table 4.3: Answer quality on **twowiki**. Exact match and token-level F1 follow the official HotpotQA evaluation script, including its yes/no short circuit; SP-EM and SP-F1 are set metrics over  $(title, sent\_id)$  supporting-fact pairs.  $n$  is the number of questions scored. No **hybrid\_rerank** run is available, so no significance marks are shown. - marks a configuration that has not been run.

| System                              | $n$ | EM | F1 | F1 95% CI | SP-EM | SP-F1 | $\Delta F1$ vs. ref. [95% CI] |
|-------------------------------------|-----|----|----|-----------|-------|-------|-------------------------------|
| <i>Baselines (weakest first)</i>    |     |    |    |           |       |       |                               |
| <b>bm25_only</b>                    | -   | -  | -  | -         | -     | -     | -                             |
| <b>dense_only</b>                   | -   | -  | -  | -         | -     | -     | -                             |
| <b>hybrid_rerank</b>                | -   | -  | -  | -         | -     | -     | -                             |
| <b>naive_rag</b>                    | -   | -  | -  | -         | -     | -     | -                             |
| <b>self_ask</b>                     | -   | -  | -  | -         | -     | -     | -                             |
| <i>Agentic system and ablations</i> |     |    |    |           |       |       |                               |
| <b>agentic_full</b>                 | -   | -  | -  | -         | -     | -     | -                             |
| <b>agentic_no_planner</b>           | -   | -  | -  | -         | -     | -     | -                             |
| <b>agentic_no_kg</b>                | -   | -  | -  | -         | -     | -     | -                             |
| <b>agentic_no_verifier</b>          | -   | -  | -  | -         | -     | -     | -                             |

Table 4.4: Retrieval quality on **twowiki**, computed with **pytrec\_eval** over the reciprocal-rank fusion of every sub-query’s ranking and judged against the gold supporting-fact documents. A multi-hop system issues several queries, so “the ranking” is not directly observed; fusing them is how the orchestrator itself pools evidence, and it needs no score calibration between BM25 and cosine. No **hybrid\_rerank** run is available, so no significance marks are shown. - marks a configuration that has not been run.

| System                              | R@2 | R@5 | R@10 | nDCG@10 | nDCG@10 95% CI | MRR |
|-------------------------------------|-----|-----|------|---------|----------------|-----|
| <i>Baselines (weakest first)</i>    |     |     |      |         |                |     |
| <b>bm25_only</b>                    | -   | -   | -    | -       | -              | -   |
| <b>dense_only</b>                   | -   | -   | -    | -       | -              | -   |
| <b>hybrid_rerank</b>                | -   | -   | -    | -       | -              | -   |
| <b>naive_rag</b>                    | -   | -   | -    | -       | -              | -   |
| <b>self_ask</b>                     | -   | -   | -    | -       | -              | -   |
| <i>Agentic system and ablations</i> |     |     |      |         |                |     |
| <b>agentic_full</b>                 | -   | -   | -    | -       | -              | -   |
| <b>agentic_no_planner</b>           | -   | -   | -    | -       | -              | -   |
| <b>agentic_no_kg</b>                | -   | -   | -    | -       | -              | -   |
| <b>agentic_no_verifier</b>          | -   | -   | -    | -       | -              | -   |

Table 4.5: Agent-specific cost on **hotpotqa**, per question. *LLM calls* counts logical calls, cache hits included; *Saved* counts calls a deterministic rule replaced; *Re-plan rate* is the fraction of questions that triggered at least one re-plan, not the mean count; *Cite grounding* is averaged only over questions that produced a non-empty answer, so it cannot be inflated by abstentions. Latency is only comparable between cold-cache runs. - marks a configuration that has not been run.

| System                              | LLM calls | Saved | Tool calls | Latency (s) | Plan depth | Sub-queries | Re-plan rate | Cite grounding |
|-------------------------------------|-----------|-------|------------|-------------|------------|-------------|--------------|----------------|
| <i>Baselines (weakest first)</i>    |           |       |            |             |            |             |              |                |
| bm25_only                           | -         | -     | -          | -           | -          | -           | -            | -              |
| dense_only                          | -         | -     | -          | -           | -          | -           | -            | -              |
| hybrid_rerank                       | -         | -     | -          | -           | -          | -           | -            | -              |
| naive_rag                           | -         | -     | -          | -           | -          | -           | -            | -              |
| self_ask                            | -         | -     | -          | -           | -          | -           | -            | -              |
| <i>Agentic system and ablations</i> |           |       |            |             |            |             |              |                |
| agentic_full                        | 3.20      | 6.20  | 12.00      | 38.7        | 1.80       | 2.00        | 0.200        | 1.000          |
| agentic_no_planner                  | -         | -     | -          | -           | -          | -           | -            | -              |
| agentic_no_kg                       | -         | -     | -          | -           | -          | -           | -            | -              |
| agentic_no_verifier                 | -         | -     | -          | -           | -          | -           | -            | -              |

Table 4.6: Agent-specific cost on **twowiki**, per question. *LLM calls* counts logical calls, cache hits included; *Saved* counts calls a deterministic rule replaced; *Re-plan rate* is the fraction of questions that triggered at least one re-plan, not the mean count; *Cite grounding* is averaged only over questions that produced a non-empty answer, so it cannot be inflated by abstentions. Latency is only comparable between cold-cache runs. - marks a configuration that has not been run.

| System                              | LLM calls | Saved | Tool calls | Latency (s) | Plan depth | Sub-queries | Re-plan rate | Cite grounding |
|-------------------------------------|-----------|-------|------------|-------------|------------|-------------|--------------|----------------|
| <i>Baselines (weakest first)</i>    |           |       |            |             |            |             |              |                |
| bm25_only                           | -         | -     | -          | -           | -          | -           | -            | -              |
| dense_only                          | -         | -     | -          | -           | -          | -           | -            | -              |
| hybrid_rerank                       | -         | -     | -          | -           | -          | -           | -            | -              |
| naive_rag                           | -         | -     | -          | -           | -          | -           | -            | -              |
| self_ask                            | -         | -     | -          | -           | -          | -           | -            | -              |
| <i>Agentic system and ablations</i> |           |       |            |             |            |             |              |                |
| agentic_full                        | -         | -     | -          | -           | -          | -           | -            | -              |
| agentic_no_planner                  | -         | -     | -          | -           | -          | -           | -            | -              |
| agentic_no_kg                       | -         | -     | -          | -           | -          | -           | -            | -              |
| agentic_no_verifier                 | -         | -     | -          | -           | -          | -           | -            | -              |

Table 4.7: Ablation study on **hotpotqa**. Deltas are measured against **agentic\_full** ( $\dagger$ ), so a *negative*  $\Delta$ F1 means the removed component was helping. **Bold** marks an ablation significantly *better* than the full system and  $\downarrow$  one significantly worse (paired bootstrap, 95% CI of the difference excluding zero); unmarked differences are not significant. **hybrid\_rerank** is repeated from the main results as the non-agentic floor, since removing planning, synthesis and verification reduces the pipeline to it. - marks a configuration that has not been run.

| System                                     | <i>n</i> | EM              | F1              | $\Delta$ F1 vs. full [95% CI] | <i>p</i> | LLM calls | Latency (s) |
|--------------------------------------------|----------|-----------------|-----------------|-------------------------------|----------|-----------|-------------|
| agentic_full                               | 5        | 0.400 $\dagger$ | 0.400 $\dagger$ |                               | - -      | 3.20      | 38.7        |
| agentic_no_planner                         | -        | -               | -               |                               | - -      | -         | -           |
| agentic_no_kg                              | -        | -               | -               |                               | - -      | -         | -           |
| agentic_no_verifier                        | -        | -               | -               |                               | - -      | -         | -           |
| hybrid_rerank ( <i>non-agentic floor</i> ) | -        | -               | -               |                               | - -      | -         | -           |

Table 4.8: Ablation study on `twowiki`. No `agentic_full` run is available, so no deltas or significance marks can be computed. `hybrid_rerank` is repeated from the main results as the non-agentic floor, since removing planning, synthesis and verification reduces the pipeline to it. - marks a configuration that has not been run.

| System                                                  | <i>n</i> | EM | F1 | $\Delta$ F1 vs. full [95% CI] | <i>p</i> | LLM calls | Latency (s) |
|---------------------------------------------------------|----------|----|----|-------------------------------|----------|-----------|-------------|
| <code>agentic_full</code>                               | –        | –  | –  | –                             | –        | –         | –           |
| <code>agentic_no_planner</code>                         | –        | –  | –  | –                             | –        | –         | –           |
| <code>agentic_no_kg</code>                              | –        | –  | –  | –                             | –        | –         | –           |
| <code>agentic_no_verifier</code>                        | –        | –  | –  | –                             | –        | –         | –           |
| <code>hybrid_rerank</code> ( <i>non-agentic floor</i> ) | –        | –  | –  | –                             | –        | –         | –           |

Table 4.9: Failure profile on `hotpotqa`. Labels are assigned post hoc by `eval/error_analysis.py` from the trace and the gold supporting facts – deterministically, first match wins – in a cascade from “the system never had a chance” to “it had everything it needed and still got it wrong”. `verifier_false_accept` and `verifier_false_reject` are counted again in the totals, independently of the cascade: the ordered label makes a false reject nearly unreachable, and it is the count that makes the Verifier→Planner loop falsifiable. Corpus titles were loaded, so the `decomposition_error` rule’s “gold facts exist in the corpus” clause is checked rather than assumed. A configuration with no run has no column.

| Label                                                 | <code>agentic_full</code> |
|-------------------------------------------------------|---------------------------|
| <i>Failure labels: count (share of all questions)</i> |                           |
| <code>parse_failure</code>                            | 0 (0.000)                 |
| <code>budget_exhausted</code>                         | 0 (0.000)                 |
| <code>retrieval_miss</code>                           | 0 (0.000)                 |
| <code>decomposition_error</code>                      | 0 (0.000)                 |
| <code>bridge_link_failure</code>                      | 0 (0.000)                 |
| <code>synthesis_error</code>                          | 1 (0.200)                 |
| <code>verifier_false_accept</code>                    | 2 (0.400)                 |
| <code>verifier_false_reject</code>                    | 0 (0.000)                 |
| <i>Totals</i>                                         |                           |
| Questions scored                                      | 5                         |
| Correct                                               | 2                         |
| Accuracy                                              | 0.400                     |
| Errors                                                | 3                         |
| Errors with no rule firing                            | 0                         |
| Verifier false accepts                                | 3                         |
| Verifier false rejects                                | 0                         |
| Recoverable (some cycle was right)                    | 0                         |

Table 4.10: Failure profile on `twowiki`. Labels are assigned post hoc by `eval/error_analysis.py` from the trace and the gold supporting facts – deterministically, first match wins – in a cascade from “the system never had a chance” to “it had everything it needed and still got it wrong”. `verifier_false_accept` and `verifier_false_reject` are counted again in the totals, independently of the cascade: the ordered label makes a false reject nearly unreachable, and it is the count that makes the Verifier→Planner loop falsifiable. Corpus titles were not loaded, so the `decomposition_error` rule assumes gold facts exist in the corpus – the permissive direction, which can blame the decomposition for a genuine corpus gap. A configuration with no run has no column.

| Label                                                 | – |
|-------------------------------------------------------|---|
| <i>Failure labels: count (share of all questions)</i> |   |
| <code>parse_failure</code>                            | – |
| <code>budget_exhausted</code>                         | – |
| <code>retrieval_miss</code>                           | – |
| <code>decomposition_error</code>                      | – |
| <code>bridge_link_failure</code>                      | – |
| <code>synthesis_error</code>                          | – |
| <code>verifier_false_accept</code>                    | – |
| <code>verifier_false_reject</code>                    | – |
| <i>Totals</i>                                         |   |
| Questions scored                                      | – |
| Correct                                               | – |
| Accuracy                                              | – |
| Errors                                                | – |
| Errors with no rule firing                            | – |
| Verifier false accepts                                | – |
| Verifier false rejects                                | – |
| Recoverable (some cycle was right)                    | – |

## Chapter 5

# Conclusions and Future Work

5.1 Summary of Contributions

5.2 Empirical Findings

5.3 Limitations

5.4 Future Work

5.5 Toward a Thesis Extension

# Bibliography